

ENSEMBLING CONVOLUTIONAL AND TRANSFORMER ARCHITECTURES FOR RETINOPATHY OF PREMATURITY SCREENING IN NEONATAL FUNDUS IMAGES

Hanish Acharla

Mountain House High School
1090 S Central Pkwy, California, United States

ABSTRACT

Neonatal eye care faces significant challenges due to diseases like Retinopathy of Prematurity (ROP). Accurate and punctual detection of ROP is crucial for the preservation of infant eyesight, but traditional screening procedures rely on manual examination bottle-necked by the limited availability of trained specialists. This paper proposes an approach to infant retinal disease detection by leveraging Convolutional Neural Networks (CNN) & Vision Transformers (ViT). Specifically, we utilize MobileNet-V2, a lightweight deep convolutional network architecture that prioritizes low computational cost, and ResNet-50, a widely adopted deep learning network that enables deeper architectures. In addition, we employ SWIN-B, a hierarchical vision transformer that uses self-attention-based shifted windows. We further develop an ensemble model where CNNs capture fine-grained vascular patterns that complement the global structural context extracted by ViTs. Our dataset is sourced from Nature Scientific and contained 6,004 images (224x224) across 13 classes which we condense to 3 to prioritize broad initial classification of ROP. Our hybrid ensemble SWIN-B and ResNet-50 model achieves an accuracy of 84% with a high degree of precision (81%) and recall (87%) that prove promising for initial screening procedures. The research findings of this study highlight the implications of deep learning ensemble CNN and ViT models in screening performance, while maintaining interoperability through class activation analysis.

Index Terms— Retinopathy of Prematurity, Biomedical image analysis, ensemble learning, convolutional neural networks, vision transformers

1. INTRODUCTION

Retinopathy of Prematurity (ROP) is a retinal disease primarily affecting infants born before 37 weeks of pregnancy. It is caused by a vasoproliferative disorder, in which the abnormal growth of blood vessels can lead to retinal lesions. While significant developments in perinatal care have reduced many complications of prematurity, ROP remains the leading cause of infantile blindness worldwide. Addressing ROP and Retinal Diseases promptly in the early period of neonatal care is crucial

for sustaining long-term eye health and quality of life development.

Recently, systematic analyses have characterized ROP with a pooled prevalence of 31.9% and that of severe ROP at 7.5% in premature infants [1]. Machine learning has become a popular approach for automated disease identification in bio-medical imaging as current methods are labor intensive and difficult to scale with the limited availability of specialists. This paper provides an ensemble deep-learning approach that focuses on automated retinal disease screening, rather than fine-grained disease staging. The primary motivation for our ensemble architecture is to combine localized feature extraction prevalent in CNNs with the global contextual modeling offered by attention mechanisms in ViTs, allowing the model to analyze both micro-vascular abnormalities and broader structural patterns.

2. RELATED WORK

Recent improvements in Machine Learning have transformed multiple industries, including biomedical imaging. Many approaches have been used for classifying retinal disease in infants and adults, with CNNs being the most profound.

Wallace et al. (2007) was one of the first to pilot a semiautomated approach to ROP classification. Their software 'ROP-tool' used handcrafted measurements of tortuosity and dilation to set a numerical baseline for disease identification. While results are limited, their work showcased the potential of automation in medical imaging [2].

Wang et al. (2022) developed a hybrid deep learning architecture composed of a custom ResNet-50 architecture that achieved a high sensitivity (100%), but with a skewed dataset of 74.8% of images consisting of No-ROP. [3].

Ramachandran et al. (2021) employed a deep learning framework for the detection of diseases on fundus images in preterm infants. A YoloV3 adapted architecture is used with a custom DarkNet-53 backbone that achieved a high accuracy, but with only 288 images in the dataset, this result may be influenced by overfitting [4].

Chen et al. (2021) utilized a deep learning model to classify Stages 1-3 of ROP and understand how models could generalize to datasets from other regions. The data was fed into a pre-trained ResNet-152, but the Area Under Precision Recall Curve

(AUPRC) results show that the model may have overfitted to data found in a specific region [5].

Luo et al (2024) combined a custom VGG-SWIN Transformer model for ROP detection that achieved optimal classification results of 90.3% AUC on a dataset consisting of 814 ROP images and 7,091 regular images. The skewed dataset adds scrutiny to the results as the data is imbalanced even with the extensive augmentation performed. The models are only trained on 2 classes and may struggle with differentiating other retinal diseases, but these experiments are not conducted [6].

Although previous work demonstrate promising results with either CNNs or ViTs, limitations arise in dataset diversity, model generalization, and feature representation. CNNs are powerful feature extractors that may miss broader structural regions, whereas transformer based models excel at capturing global context. These limitations motivate the development of a hybrid deep learning ensemble architecture to improve robustness in early ROP screening.

3. MATERIALS AND METHODS

3.1. Dataset

The dataset used in this study was sourced from Nature Scientific Datasets, where “Retinal Image Dataset of Infants and ROP” was selected for its robust balance between dataset size and recency [7]. The dataset included 6,004 fundus images (size=224,224) split across the 13 classes displayed in Table 1. The primary objective of this study is to provide broad initial screening of ROP cases and in order to improve model stability, we exclude rare classes (Hamartomas, Hypoplasia n.II, Toxoplasma) and combine all ROP sub-classes into a single class and retain the physiological & hemorrhage classes. Our processed data set included physiological (2,980 images), ROP (1,729 images), and hemorrhage (1,061 images). We split the dataset into the training set (70% or 4,039 images), validation (15% or 866 images) and test set (15% or 865 images) at the patient level.

3.2. Preprocessing and Augmentation

We apply Contrast Limited Adaptive Histogram Equalization (CLAHE), a localized contrast-enhancement tool to improve the visibility of features in an image [8]. We removed the black border surrounding the circular fundus images in the original dataset to limit irrelevant feature learning. To expand our dataset and enhance model robustness, we implement a series of augmentations on our training dataset. This includes Rotation, Horizontal flip, ShiftScaleRotate, RandomBrightnessContrast, GaussianBlur, and GaussNoise each at varying probabilities to improve robustness in image variability. To balance the underrepresented classes, we separately sample images from each class till ROP, hemorrhage, and physiological in the train set had 2,500 images each. Our final train, validation, and test dataset consisted of 7,500, 1,015, and 1,100 images respectively. Augmentation was only applied to the training set after splitting.

Table 1: Disease Classes in Original Dataset

Diagnosis	Description
ROP 0	No clear demarcation line
ROP 1	Demarcation line only.
ROP 2	Raised ridge at demarcation.
ROP 3	Extraretinal fibrovascular proliferation.
ROP 4	Partial retinal detachment.
ROP 5	Total retinal detachment.
ROP 4A	Subtotal detachment sparing fovea.
ROP 4B	Subtotal detachment involving fovea.
A-ROP	Rapid posterior progression.
St.p-ROP	Sight-threatening disease.
Hamartomas	Benign retinal tissue overgrowth.
Hemorrhage	Retinal or vitreous bleeding.
Hypoplasia n. II	Underdeveloped optic nerve.
Toxoplasma	Necrotizing retinal inflammation.
Physiological	Healthy retinal fundus

3.3. Methodology

We first employ MobileNetV2 [9], known for its compact size and utility in embedded systems. It achieves this by utilizing depthwise separable convolutions that a single filter per input channel and combine the channels with a 1×1 convolution. Additionally, inverted residuals are used for efficient gradient flow, and linear bottleneck layers are used to preserve essential features. Combining these processes enables MobileNet-V2 to excel with relatively low compute.

Next, ResNet-50 [10] is a 50-layer CNN that uses residual learning to train models. It utilizes bottleneck residual blocks that enable shortcut connections that can either add the input directly or apply a projection to match the dimension size using 1×1 convolutions. This process allows the network to learn residual functions (difference from the desired output and input) compared to entire feature maps. He et al. (2016), in their famous paper “Deep Residual Learning for Image Recognition” describe the improvements residual learning has on model performance and overcoming the vanishing gradient problem, paving the way for deeper models like ResNet-152. Coupled with batch normalization and ReLU activations, ResNet-50 provides powerful feature representations that deliver high performance in image classification.

The ViT we use in this study is SWIN-Base (SWIN-B) [11], which builds hierarchical feature maps similar to CNNs but uses window-based self-attention instead of global self-attention (a common approach in other ViTs). Windows in each layer do not overlap, so to gain cross-window interactions without quadratic costs, consecutive layers shift window partitions so neighboring windows overlap over time. This gives SWIN-Transformers their hierarchical vision capabilities in linear complexity. Position bias encodes attention preference based on relative position, and residual connections enable stable training. SWIN-Transformers come in multiple sizes (SWIN-Tiny/Small/Base/Large) that attend to the number of

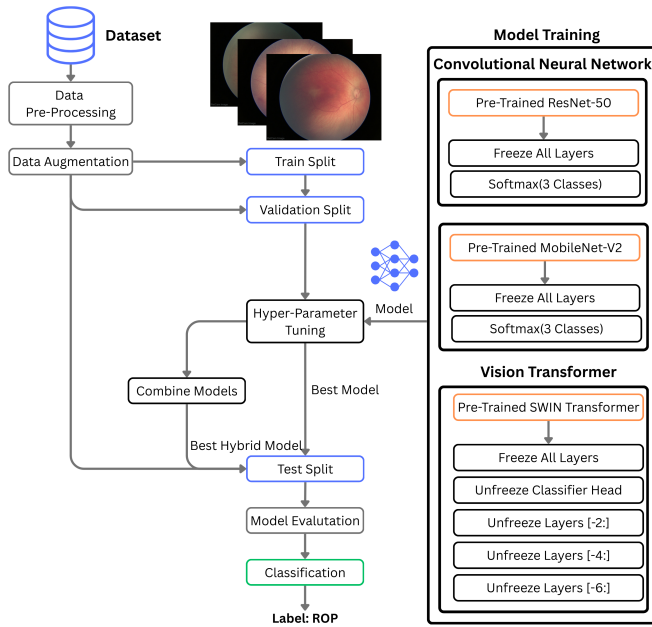


Fig. 1: Block Diagram of Proposed Methodology

layers and complexity of the model. SWIN-B offers strong performance on classification tasks with relatively less compute compared to other ViTs.

While MobileNet-V2 is strong in limited computational environments, the lower parameter count often results in degraded performance compared to larger models. ResNet-50 overcomes some of the performance issues, but as a result, training takes much longer. SWIN-B has nearly 4x parameters as ResNet-50, but its increased stability and accuracy can be necessary.

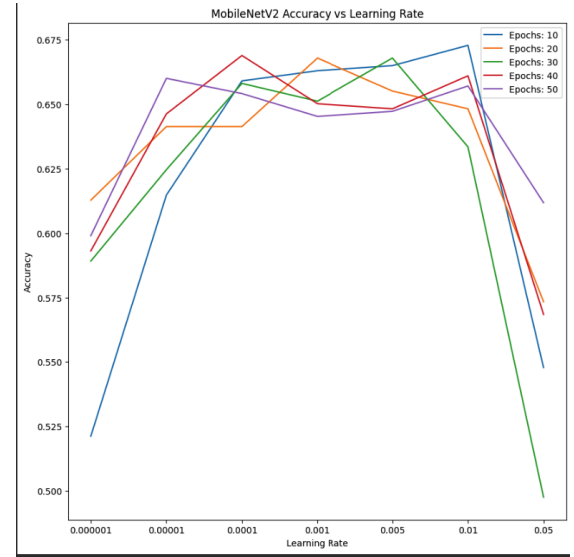
All experiments were conducted using Google Colab A100 GPU runtime environment with Tensorflow as the deep learning framework for model training. This configuration proved useful when accommodating the higher parameter count of the Transformer models.

4. RESULTS AND DISCUSSION

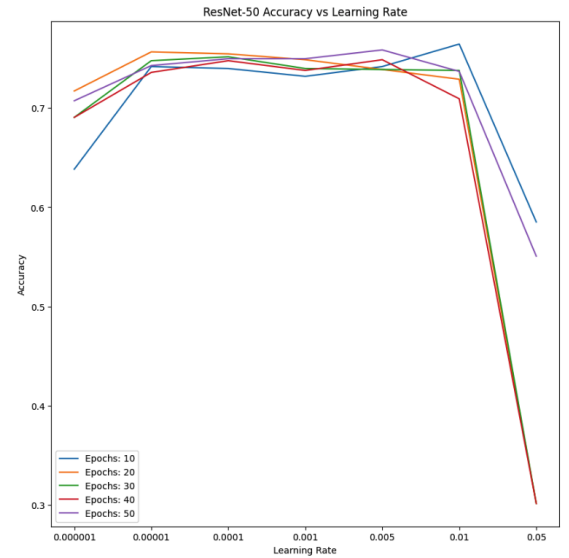
4.1. Quantative Results

ImageNet transfer learning [12] is used for CNN and ViT approaches with a fully connected classification layer (Softmax activation) for output predictions. All layers in the CNNs are frozen to retain learned weights from the pre-training, while we tune the final classification layer. We tune the hyperparameters with multiple learning rates ($learning_rate = 0.05, 0.01, 0.005, 0.001, 0.0001, 0.00001, 0.000001$) and epochs ($epoch = 10, 20, \dots, 50$) Batch size of 32 is used for all models and early stopping criteria halts the training process when limited improvements were made on the validation dataset to reduce overfitting.

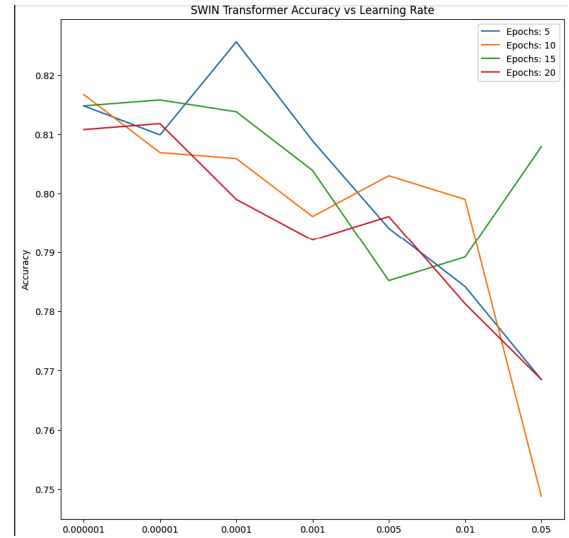
Figure 2(a) showcases the results of MobileNet-V2 hyper-



(a) MobileNet-V2



(b) ResNet-50



(c) SWIN-B

Fig. 2: Hyper Parameter Tuning of CNN's & ViT

parameter tuning, and the large variation between accuracies cements the importance of testing the different combinations of hyperparameter values. A clear trend that can be observed across all epochs is that the model performs poorly at the highest and lowest learning rates of 0.05 and 0.000001, likely caused by the model overshooting/not effectively progressing towards the optimal weights. Figure 2(b) displays a similar MPT for ResNet-50 and while this model performs better overall, it faces a similar issue as our MobileNet model, with the highest learning rate of 0.05 struggling to converge. A positive result of ResNet-50 is that the model performs relatively well at the lowest learning rate of 0.000001, likely caused by the model’s ability to extract deeper features.

For SWIN-B, we reduce the number of epochs because of its higher parameter count and use a slightly different transfer-learning approach. We began with training our classification head with a variable number of epochs($epochs = \{5, 10, 15, 20\}$), and gradually unfreeze more layers. The process begins with training the last 2 layers for 20 epochs, subsequently followed by the final 4 and final 6 layers for 50 epochs each. Figure 2(c) displays the results of our training procedure and the obvious pattern is that the model decreases in performs as the learning rate increases. SWIN-B utilizes transformer-style attention with more parameters than the CNN’s used in this study, increasing sensitivity to large learning rates. SWIN-B has the best result of 0.82 accuracy in the validation dataset with 10 epochs in the classifier head and a learning rate of 0.0001.

The 4 metrics used to assess the models on the test set were classification accuracy, precision, recall, and the F-1 score, which work together to highlight where the models may be struggling. All metrics reported are macro-averages as there is not a strong imbalance that must be corrected for. Recall is particularly important as high recall ensures diseased cases are correctly identified with a lower degree of False Negatives. We construct a ResNet-50 + SWIN-B ensemble where the models individually output class confidence scores that are aggregated with an equal weight-age (0.5,0.5) for final class prediction. Tuning the weight-age to favor a certain model(Ex. Prioritizing ResNet-50 (0.6,0.4)) degraded the model performance. Table II presents the results for the individual models and the ensemble model. MobileNet-V2 performs poorly on the dataset with a 42% accuracy and similar precision and recall scores, but this is heavily due to the model’s low parameter count and focus on computational efficiency. ResNet-50 achieves an accuracy of 68% on the test set, a sharp increase from MobileNet-V2, without any blatant issues elsewhere. SWIN-B performs the best as a stand-alone model, garnering an accuracy of 77% high with precision (0.74), recall (0.79), and F-1 score 0.75). These results show the true power of ViT’s in image classification as the hierarchal feature maps drawn from multi-head attention, sharply increase performance. Combining ResNet-50 and SWIN-B for ensemble predictions provides the best results with 84% accuracy and the highest precision, recall, and F-1 scores of 0.81, 0.87, and 0.82 respectively. The ensemble model’s

Table 2: Summary results of Deep CNN and ViT Models on test data

Model	Acc.	Prec.	Recall	F1
MobileNet-V2 (Epochs=10, LR=0.01)	0.42	0.46	0.50	0.40
ResNet-50 (Epochs=10, LR=0.01)	0.68	0.69	0.73	0.66
SWIN-B (Epochs=10, LR=0.01)	0.77	0.74	0.79	0.75
SWIN-B + ResNet-50 (Ensemble)	0.84	0.81	0.87	0.82

high recall of 87% is particularly important for screening applications where missing diseased cases can lead to irreversible vision loss shortly after, whereas false positives associated with precision simply lead to further specialist examination to confirm the diagnoses. Thus, the high recall achieved by the ensemble model supports its role in early screening procedures.

Figure 3 displays the promising results of the models through Normalized Confusion Matrices (NCM) ResNet-50 garners high performance across all classes, particularly excelling in ROP with 83.02% accuracy. Issues arise with hemorrhage classification as 46.29% of this class is misclassified. While SWIN-B performs similarly to ResNet-50 in the ROP and physiological class, it resolves the issues in the hemorrhage classes with an updated 18.87% misclassification rate. When ensembling models, we decided to not include MobileNet-V2 as it’s low performance may have been detrimental to the results. Our ResNet-50 + SWIN-B NCM in Figure 3(c) shows promising accuracy gains compared to individual models with 91.19% of hemorrhage images correctly classified. One explanation for this improvement is CNNs may overlook the broader structural pattern associated with hemorrhages that the Transformer’s self-attention mechanisms account for, enabling the model to better distinguish the overall region. The ensemble model continues to reduce False positives accurately, identifying 92% of physiological images.

4.2. Discussion

We conduct an ablation study using ResNet-50 as a baseline to evaluate the effects of data preprocessing and augmentation, as summarized in Table III. The results highlight the effectiveness of data-augmentation in providing the primary performance gains across all configurations. Pairing augmentation with additional pre-processing like CLAHE and border removal further improved classification performance, seen in the full ResNet-50 pipeline(Table II) where a combination of all 3 processing stages is used. In contrast, applying heavy pre-processing without augmentation degrades performance, implying that augmentation is important for robust generalization in screening procedures.

Grad-CAM (Gradient-weighted Class Activation Mapping) is a technique that generates heatmaps to visualize the specific areas of an image that drive classification decisions. Using ResNet-50, we generated Grad-CAMs for each class (physiological, ROP, and hemorrhage) as displayed in Figure 4. Phys-

Table 3: Effect of preprocessing and augmentation using a ResNet-50 backbone with test data

Setting	Acc.	Prec.	Recall	F1
BASE + Aug	0.66	0.66	0.71	0.65
CLAHE + Aug	0.63	0.64	0.66	0.62
Border + Aug	0.63	0.64	0.66	0.62
CLAHE + Border (No Aug)	0.53	0.60	0.63	0.52

iological images portray a diffuse activation across the image that is centered on healthy retinal areas, reflecting global structural consistency. ROP has a strong localized attention on the posterior pole and the surrounding abnormal vascular region, consistent with vascular changes associated with ROP. Hemorrhage cases contain high activation around bleeding regions. These class activation mappings suggest that our models rely on meaningful features rather than irrelevant background artifacts.

Limitations of our study include rare disease classes such as hamartomas, optic nerve hypoplasia, and toxoplasma being excluded from the dataset to improve class balance which may limit model generalization. Future work points towards the inclusion of rare disease categories and multi-institutional datasets to combat regional overfitting.

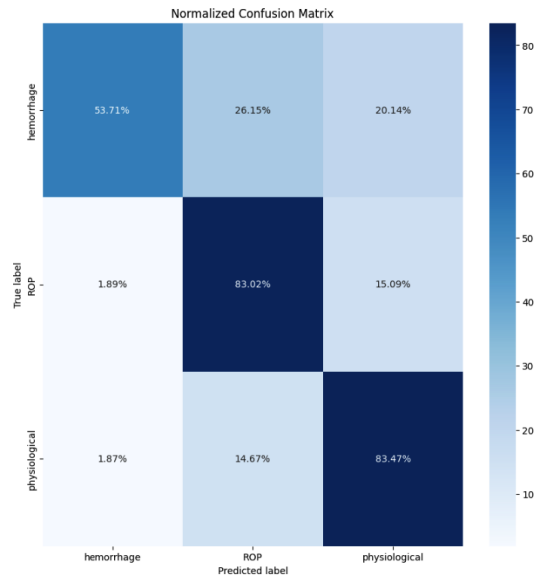
5. CONCLUSION

This study combines vision transformers (SWIN-B) with convolution neural networks (ResNet-50) to improve performance in automated screening of retinal diseases prevalent in neonatal fundus images. The 84% accuracy of the ensemble model in the test dataset indicates that the hierarchical feature maps in SWIN-B complement the residual learning from ResNet-50, leading to efficient and accurate predictions. Utilizing CLAHE normalization, border removal, and image augmentation techniques improves class balance and data robustness. Ablation analysis highlights the critical role of augmentation in classification results and the complementary improvements pre-processing has when combined appropriately. Grad-CAM visualizations provide class-specific activation patterns that align with clinically important retinal features, supporting model interpretability. Overall, the clear performance gains from ensembling vision transformers with deep convolutional neural networks demonstrate effectiveness for retinopathy of prematurity screening and may be useful in generalizing to other medical imaging tasks.

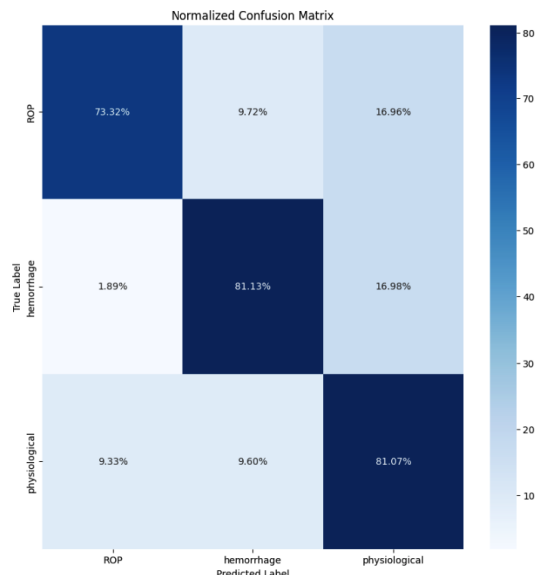
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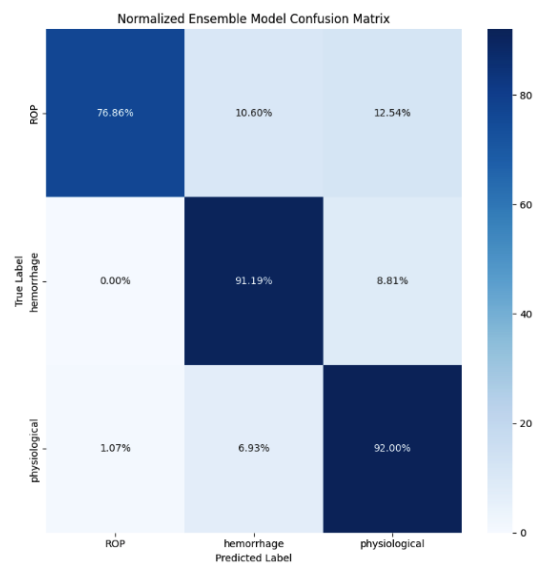
Fig. 3: Comparison of normalized confusion matrices.



(a) ResNet-50



(b) SWIN-B



(c) ResNet-50 + SWIN-B

Fig. 4: Grad-CAM visualizations (ResNet-50) for representative fundus images from each class. Left: original image. Right: Grad-CAM overlay.

